

Optimizing Investment Portfolios Using Machine Learning: A Case Study on the Moroccan Market

Paper Abstract:

This paper introduces the Moroccan Market Adaptive Transformer (MMAT), a novel adaptation of Transformer architecture for stock price prediction in the Moroccan market, addressing critical research gaps in emerging markets. Our innovation lies in tailoring the Transformer architecture specifically for the unique characteristics of emerging markets through our Emerging Market Preprocessing Framework (EMPF) and Market-Adaptive Attention Mechanism (MAAM). We implement this comprehensive model on ten major Moroccan companies across diverse sectors and evaluate its performance using a Multi-Metric Evaluation Framework. Our research demonstrates the model's exceptional predictive capabilities, achieving an average Mean Absolute Error (MAE) of 0.0880, Root Mean Squared Error (RMSE) of 0.1180, and R^2 score of 0.9145 across the tested companies. Unlike previous studies that primarily focused on developed markets or single indices, our work pioneers the application of attention-based models to individual Moroccan stocks, with rigorous cross-sectoral analysis revealing significant performance variations that provide actionable investment insights. Our ablation studies confirm that our market-specific adaptations substantially improve performance over standard implementations. These findings contribute valuable methodological and practical insights for investors and researchers interested in emerging markets, demonstrating that properly adapted deep learning architectures can overcome the challenges posed by market inefficiencies and data limitations unique to developing economies.